

Hong Yi Huang

☎ +44 (0)7579956791 or +36 304167944

✉ 11612012@mail.sustech.edu.cn

🌐 hongyihuang328.github.io

Alfréd Rényi Institute of Mathematics, Reáltanoda utca 13-15., H-1053, Budapest, Hungary

Research interests

- Group theory (permutation groups, simple groups)
- Algebraic combinatorics

Employment

Research Fellow, HUN-REN Alfréd Rényi Institute of Mathematics, Hungary *from May 2026*

- Academic host: Professor László Pyber

Research Scholar, Southern University of Science and Technology (SUSTech), China *Nov 2025 – Apr 2026*

- Academic host: Professor Cai Heng Li

Research Fellow, University of St Andrews, United Kingdom *Apr 2025 – Oct 2025*

- Academic host: Professor Colva Roney-Dougal
- With an Early Career Fellowship from the London Mathematical Society (LMS)
- Financial support: LMS & Heilbronn Institute for Mathematical Research

Education

University of Bristol (UoB), United Kingdom *Jan 2021 – Mar 2025*

- PhD in Mathematics
- Supervisor: Professor Tim Burness
- Funding: China Scholarship Council
- Thesis: *Bases for permutation groups and related problems*

SUSTech, China *Sept 2016 – July 2020*

- Bachelor of Science
- Major: Mathematics and Applied Mathematics
- Supervisor: Professor Cai Heng Li
- Thesis: *On valency problems of Saxl graphs of almost simple primitive groups with soluble stabiliser*

Distinctions

LMS Early Career Fellowship (6 fellows in 2025) *2025*

Oberwolfach Leibniz Graduate Students grant, Mathematical Research Institute Oberwolfach *2025*

Alumni Grant: Conference Travel Fund (£300), UoB *2024*

China Scholarship Council (full PhD scholarship at UoB) *2021–25*

Last updated: 27 July 2026

Journal publications and preprints

- [10] H.Y. Huang and C.M. Roney-Dougal, *Greedy bases and relational complexity of diagonal type groups*, submitted (2026). arxiv:2605.16032
- [9] S.D. Freedman, H.Y. Huang, M. Lee and K. Rekvényi, *On the generalised Saxl graphs of permutation groups*, **Algebr. Comb.** 9 (2026), 611–648. doi:10.5802/alco.493
- [8] T.C. Burness and H.Y. Huang, *On the intersections of nilpotent subgroups in simple groups*, **Proc. London Math. Soc.** 132 (2026), Paper No. e70134, 48 pp. doi:10.1112/plms.70134
- [7] T.C. Burness and H.Y. Huang, *On the intersections of Sylow subgroups in almost simple groups*, **J. Algebra** 690 (2026), 596–631. doi:10.1016/j.jalgebra.2025.11.003
- [6] H.Y. Huang, C.H. Li and Y.L. Xie, *Fixers and derangements of finite permutation groups*, **J. Algebraic Combin.** 61 (2025), Paper No. 57, 44 pp. doi:10.1007/s10801-025-01425-3
- [5] H.Y. Huang, C.H. Li and Y.Z. Zhu, *On finite permutation groups of rank three*, **J. Algebra** 666 (2025), 703–732. doi:10.1016/j.jalgebra.2024.11.028
- [4] H.Y. Huang, *Base sizes of primitive groups of diagonal type*, **Forum Math. Sigma** 12 (2024), Paper No. e2, 43 pp. doi:10.1017/fms.2023.121
- [3] T.C. Burness and H.Y. Huang, *On base sizes for primitive groups of product type*, **J. Pure Appl. Algebra** 227 (2023), Paper No. 107228, 43 pp. doi:10.1016/j.jpaa.2022.107228
- [2] T.C. Burness and H.Y. Huang, *On the Saxl graphs of primitive groups with soluble stabilisers*, **Algebr. Comb.** 5 (2022), 1053–1087. doi:10.5802/alco.238
- [1] J. Chen and H.Y. Huang, *On valency problems of Saxl graphs*, **J. Group Theory** 25 (2022), 543–577. doi:10.1515/jgth-2021-0123

Conference & seminar organisation

26th Postgraduate Group Theory Conference (PGTC), UoB

July 2025

- With Marina Anagnostopoulou-Merkouri, Pénélope Azuelos, David Guo, Merlin Haith Rowlett and Sarah Martin
- Funded by the London Mathematical Society, the Heilbronn Institute for Mathematical Research, Jane Street Capital, and the School of Mathematics of UoB
- Website: <https://sites.google.com/view/pgtc2025>

Bristol Junior Algebra and Combinatorics Seminar (BJACS), UoB

Feb–May 2024

- Founding co-organiser
- With Marina Anagnostopoulou-Merkouri and Merlin Haith Rowlett
- Website: <https://sites.google.com/view/bjacs>

4th International Conference on Groups, Graphs and Combinatorics, SUSTech

Nov 2023

- With a group of people led by Cai Heng Li

Seminars on Groups and Graphs, SUSTech (online)

2021–24

- Founding co-organiser
- With Zhaochen Ding (2021–24), Yilin Xie (2021–23), Junyan Chen (2024) and Fu-Gang Yin (2024)
- Website: <https://www.gtseminar.xyz/>

Conference invitations

<i>Algebraic groups, geometry, invariants and related topics</i> , University of Essex	July 2026
<i>Finite Groups, Fusion Systems and Applications</i> , Mathematical Research Institute Oberwolfach	Mar 2025
<i>Young Group Theorists Workshop</i> , SwissMAP, Les Diablerets, Switzerland	Sept 2022

Invited talks & seminars

2026

<i>Simple groups, Sylow subgroups and intersections</i> , GGI 2026, University of Essex	13 July
<i>Groups, permutations and bases</i> , University of St Andrews	5 Feb

2025

<i>Intersections of Sylow subgroups in simple groups</i> , Yunnan University of Finance and Economics	23 Dec
<i>Transitive subgroups of primitive groups</i> , Groups and Combinatorics, China Three Gorge University	25 Oct
<i>Intersections of Sylow subgroups in simple groups</i> , Central China Normal University	24 Oct
<i>Intersections of Sylow subgroups in simple groups</i> , Peking University	25 Sept
<i>Intersections of Sylow subgroups in simple groups</i> , Beijing Jiaotong University	24 Sept
<i>Permutation groups, transitive subgroups and bases</i> , Pure Maths Colloquium, University of St Andrews	17 Apr
<i>Permutation groups, symmetry breaking and probability</i> , Great Bay University	26 Feb
<i>Permutation groups, symmetry breaking and probability</i> , Guangxi University	25 Feb

2024

<i>Permutation groups, symmetry breaking and probability</i> , Zhejiang University	24 Dec
<i>Bases and Saxl graphs in the past four years</i> , Discrete Mathematics Seminar, SUSTech	12 Dec
<i>Bases and Saxl graphs for permutation groups</i> , BJACS, UoB	18 Sept
<i>Bases for permutation groups</i> , University of Milano-Bicocca, Italy	6 Sept
<i>Permutations, bases and low rank groups</i> , Joint Algebra Seminar, RWTH Aachen	13 June
<i>Permutations, bases and low rank groups</i> , Advanced Seminar on Group Theory, HHU Düsseldorf	11 June
<i>Permutations, bases and low rank groups</i> , Seminar Groups and Representations, Kaiserslautern-Landau	6 June
<i>Permutations, bases and low rank groups</i> , Algebra Seminar, UoB	19 Mar
<i>Bases for permutation groups</i> , Algebra Seminar, University of Warwick	29 Jan

2023

<i>Bases for permutation groups</i> , Algebra Seminar, University of Birmingham	30 Nov
<i>Symmetry breaking on graphs and groups</i> , Seminars on Discrete Mathematics, Xiamen University	26 Sept
<i>Permutation groups: bases and Saxl graphs</i> , Capital Normal University	16 Sept
<i>Bases for permutation groups</i> , SUSTech	7 Sept
<i>Bases for permutation groups</i> , Junior London Algebra Colloquium, Imperial College London	23 Mar

2022

<i>Bases, distinguishing partitions and probabilistic methods</i> , The University of Melbourne (online)	18 Aug
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2021

<i>Groups, graphs and transitivity</i> , Junior Algebra Colloquium, UoB (online)	21 May
<i>The probabilistic method in group theory</i> , Discrete Mathematics Seminar, SUSTech	22 Apr

2020

<i>On valency problems of Saxl graphs</i> , Discrete Mathematics Seminar, SUSTech	26 Nov
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Other talks

2025

<i>Intersections of Sylow subgroups in simple groups</i> , Finite Groups and their Representations, Xiamen	18 Oct
<i>Transitive subgroups of primitive groups</i> , Mathematical Research Institute Oberwolfach, Germany	20 Mar
<i>Regular subgroups of primitive groups</i> , Groups, Graphs and Combinatorics, Kunming	12 Jan

2024

<i>The generalised Saxl graphs of finite permutation groups</i> , Topics in Group Theory, Padova, Italy	11 Sept
<i>Permutation groups of rank three</i> , Group Theory in Florence IV, Florence, Italy	2 July

2023

<i>Bases for permutation groups</i> , 16th National Conference on Algebra, Quanzhou	18 Nov
<i>Bases for permutation groups</i> , 4th International Conference on Groups, Graphs and Combinatorics	11 Nov
<i>Symmetry breaking on primitive groups</i> , Seminars on Groups and Graphs, SUSTech	4 Mar

2022

<i>Base-two primitive permutation groups</i> , Young Group Theorists Workshop, SwissMAP, Les Diablerets	5 Sept
<i>Base-two primitive permutation groups</i> , Groups St Andrews, Newcastle, UK	4 Aug
<i>Base-two primitive permutation groups and their Saxl graphs</i> , Groups and Graphs	24 July
<i>Base-two primitive permutation groups</i> , 23rd Postgraduate Group Theory Conference	8 July
<i>Bases for primitive permutation groups</i> , Group Theory Seminar, SUSTech	24 May
<i>Regular orbits of primitive groups on power sets</i> , Group Theory Seminar, SUSTech	19 Feb

2021

<i>Base-two primitive permutation groups and their Saxl graphs</i> , 43rd Australasian Combinatorics Conference	13 Dec
<i>Regular suborbits of finite primitive groups</i> , Groups, Graphs and Combinatorics	14 Nov
<i>Base-two primitive permutation groups and their Saxl graphs</i> , LMS Graduate Student Meeting	8 Nov
<i>The distinguishing number of permutation groups</i> , Group Theory Seminar, SUSTech	9 Oct
<i>On valency problems of Saxl graphs</i> , Workshop on Group Actions and Transitive Graphs	2 Jan

2020

<i>On valency problems of Saxl graphs</i> , SUSTech	17 Nov
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Other conferences

2026

AGGITatE 2026, University of Essex	July
<i>Pure maths in the age of AI</i> , University of Essex	July
<i>LMS Research School on Growth and Expansion in Groups</i> , University of Warwick	July
<i>Modern Flavours of Finite Groups</i> , University of Warwick	June

2025

<i>Conference on Algebraic Graph Theory and Combinatorics</i> , SUSTech	Nov
<i>Groups and Combinatorics</i> , China Three Gorge University, Yichang, China	Oct
<i>Finite Groups and their Representations</i> , Xiamen University, Xiamen, China	Oct
<i>Groups, Graphs and Combinatorics</i> , Yunnan University, Kunming, China	Jan

2024

<i>Topics in Group Theory</i> , University of Padova	Sept
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<i>Group Theory in Florence IV</i> , University of Florence	July
<i>New perspectives in Computational Group Theory</i> , University of Warwick	June
2023	
<i>16th National Conference on Algebra</i> , Quanzhou, China	Nov
<i>International Conference on Combinatorics and Finite Geometry</i> , SUSTech	Sept
<i>CMI-HIMR Summer School: Probabilistic Group Theory</i> , UoB	June
<i>Fry Conference Series: New Perspectives in Pure Mathematics</i> , UoB	Mar
2022	
<i>Groups St Andrews</i> , Newcastle University	Aug
<i>Simple groups, Representations and Applications</i> , Isaac Newton Institute, University of Cambridge	July
<i>Groups and Graphs</i> , Jiangxi University of Science and Technology (online)	July
<i>23rd Postgraduate Group Theory Conference</i> , London	July
2021	
<i>43rd Australasian Combinatorics Conference</i> , The University of Melbourne (online)	Dec
<i>Groups, Graphs and Combinatorics</i> , SUSTech (online)	Nov
<i>LMS Graduate Student Meeting</i> , London Mathematical Society (online)	Nov
<i>Workshop on Group Actions and Transitive Graphs</i> , Kunming, China	Jan
Before 2020	
<i>Workshop on Combinatorics and Graph Theory</i> , SUSTech	Oct 2020
<i>International Conference on Algebraic Combinatorics</i> , Jiaozuo, China	Sept 2019
<i>The Third International Conference on Group Actions and Transitive Graphs</i> , SUSTech	Oct 2018

Teaching

Tutorial leader, School of Mathematics, UoB

- 2023–2024 Teaching Block 2: MATH10010 Introduction to Group Theory (3 tutorial groups)
- 2021–2022 Teaching Block 1: MATH10010 Introduction to Proofs (3 tutorial groups)

Teaching assistant, Department of Mathematics, SUSTech

- 2019–2020 Spring: MA109 Advanced Linear Algebra (1 out of 3 tutorial groups)
- 2019–2020 Fall: MA107 Advanced Linear Algebra I (2 out of 2 tutorial groups)
- 2018–2019 Spring: MA104b Linear Algebra II (1 out of 4 tutorial groups)

Homework marker, Department of Mathematics, SUSTech

- 2020–2021 Spring: MA321 Group Representation Theory, and MA219-16 Elementary Number Theory
- 2019–2020 Fall: MA321 Group Representation Theory, and MA219 Abstract Algebra (H)
- 2018–2019 Spring: MAT8010 Combinatorics (PG)
- 2017–2018 Fall: MA219-16 Elementary Number Theory, and MA213-16 Mathematical Analysis

Research visits

2026: University of St Andrews (1 week, Colva Roney-Dougal)

2025: Yunnan University of Finance and Economics (1 week, Cixuan Wu); Central China Normal University (3 days, Yuanyang Zhou); Xiamen University (3 days, Jiyong Chen); Peking University (1 week, Jiping Zhang); Beijing Jiaotong University (1 day, Fu-Gang Yin); Warwick (1 day, Sean Eberhard and Gareth Tracey); Great

Bay University (1 day, Weicong Li); Guangxi University (1 day, Huye Chen); Yunnan University (1 week, Lei Wang)

2024: Zhejiang University (1 week, Tao Feng); SUSTech (1 week, Cai Heng Li); Milano-Bicocca (3 days, Pablo Spiga); Warwick (1 day, Gareth Tracey); RWTH Aachen (3 days, Alice Niemeyer); HHU Düsseldorf (2 days, Benjamin Klopsch); Kaiserslautern-Landau (1 week, Gunter Malle); Imperial College London (3 days, Kamilla Rekvényi)

2023: Xiamen University (1 week, Jiyong Chen); Capital Normal University (1 week, Shaofei Du); SUSTech (1 month, Cai Heng Li)

2021: SUSTech (2 months, Qing Xiang)

2020: SUSTech (6 months, Cai Heng Li)

Professional activities

- Refereeing assignments: J. Algebra, European J. Combin., J. Group Theory, J. Combin. Theory Ser. A, Proc. Roy. Soc. Edinburgh Sect. A
- Reviewer for AMS Mathematical Reviews (5 reviews) *since 2023*
- Member of the American Mathematical Society *since 2026*
- Associate Member of the London Mathematical Society *2024–25*
- Member of the Combinatorial Mathematics Society of Australasia *2022*